AI-Assisted Threat Detection Dashboard Model Research

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Abstract

This report presents an AI-Assisted Threat Detection Dashboard that combines cybersecurity data analysis with artificial intelligence. The proposed solution aggregates security logs, threat intelligence feeds and vulnerability information to identify suspicious activities. Machine learning models are evaluated to determine the most suitable algorithm for accurate threat detection. The dashboard enables security analysts to monitor attack trends, prioritize incidents and improve decision-making. This report presents an AI-Assisted Threat Detection Dashboard that combines cybersecurity data analysis with artificial intelligence. The proposed solution aggregates security logs, threat intelligence feeds and vulnerability information to identify suspicious activities. Machine learning models are evaluated to determine the most suitable algorithm for accurate threat detection. The dashboard enables security analysts to monitor attack trends, prioritize incidents and improve decision-making.

Introduction

Cybersecurity has become a critical requirement for modern organizations. Networks generate millions of security events every day from firewalls, servers, cloud services and endpoint devices. Manual investigation is time-consuming and traditional signature-based detection cannot identify every new attack. AI-based approaches learn patterns from historical data and detect abnormal behaviour more effectively. Cybersecurity has become a critical requirement for modern organizations. Networks generate millions of security events every day from firewalls, servers, cloud services and endpoint devices. Manual investigation is time-consuming and traditional signature-based detection cannot identify every new attack. AI-based approaches learn patterns from historical data and detect abnormal behaviour more effectively. Cybersecurity has become a critical requirement for modern organizations. Networks generate millions of security events every day from firewalls, servers, cloud services and endpoint devices. Manual investigation is time-consuming and traditional signature-based detection cannot identify every new attack. AI-based approaches learn patterns from historical data and detect abnormal behaviour more effectively. Cybersecurity has become a critical requirement for modern organizations. Networks generate millions of security events every day from firewalls, servers, cloud services and endpoint devices. Manual investigation is time-consuming and traditional signature-based detection cannot identify every new attack. AI-based approaches learn patterns from historical data and detect abnormal behaviour more effectively. Cybersecurity has become a critical requirement for modern organizations. Networks generate millions of security events every day from firewalls, servers, cloud services and endpoint devices. Manual investigation is time-consuming and traditional signature-based detection cannot identify every new attack. AI-based approaches learn patterns from historical data and detect abnormal behaviour more effectively.

Problem Statement

Organizations face challenges in identifying real threats from massive volumes of security events. Existing systems generate numerous alerts, making it difficult for analysts to prioritize incidents. The project aims to automate threat detection using machine learning and present results through an interactive dashboard. Organizations face challenges in identifying real threats from massive volumes of security events. Existing systems generate numerous alerts, making it difficult for analysts to prioritize incidents. The project aims to automate threat detection using machine learning and present results through an interactive dashboard. Organizations face challenges in identifying real threats from massive volumes of security events. Existing systems generate numerous alerts, making it difficult for analysts to prioritize incidents. The project aims to automate threat detection using machine learning and present results through an interactive dashboard. Organizations face challenges in identifying real threats from massive volumes of security events. Existing systems generate numerous alerts, making it difficult for analysts to prioritize incidents. The project aims to automate threat detection using machine learning and present results through an interactive dashboard.

Objectives

Develop a centralized dashboard, preprocess cybersecurity datasets, compare multiple AI models, detect anomalies, prioritize threats, visualize results and improve cybersecurity decision making. Develop a centralized dashboard, preprocess cybersecurity datasets, compare multiple AI models, detect anomalies, prioritize threats, visualize results and improve cybersecurity decision making. Develop a centralized dashboard, preprocess cybersecurity datasets, compare multiple AI models, detect anomalies, prioritize threats, visualize results and improve cybersecurity decision making.

Literature Review

Research in intrusion detection shows that ensemble learning algorithms such as Random Forest and XGBoost consistently outperform individual classifiers on structured cybersecurity datasets. Isolation Forest and Autoencoders are effective for anomaly detection, while Support Vector Machines perform well on high-dimensional data. Modern systems also integrate MITRE ATT&CK, CVE and threat intelligence feeds to enrich security events and provide context for analysts. Research in intrusion detection shows that ensemble learning algorithms such as Random Forest and XGBoost consistently outperform individual classifiers on structured cybersecurity datasets. Isolation Forest and Autoencoders are effective for anomaly detection, while Support Vector Machines perform well on high-dimensional data. Modern systems also integrate MITRE ATT&CK, CVE and threat intelligence feeds to enrich security events and provide context for analysts. Research in intrusion detection shows that ensemble learning algorithms such as Random Forest and XGBoost consistently outperform individual classifiers on structured cybersecurity datasets. Isolation Forest and Autoencoders are effective for anomaly detection, while Support Vector Machines perform well on high-dimensional data. Modern systems also integrate MITRE ATT&CK, CVE and threat intelligence feeds to enrich security events and provide context for analysts. Research in intrusion detection shows that ensemble learning algorithms such as Random Forest and XGBoost consistently outperform individual classifiers on structured cybersecurity datasets. Isolation Forest and Autoencoders are effective for anomaly detection, while Support Vector Machines perform well on high-dimensional data. Modern systems also integrate MITRE ATT&CK, CVE and threat intelligence feeds to enrich security events and provide context for analysts. Research in intrusion detection shows that ensemble learning algorithms such as Random Forest and XGBoost consistently outperform individual classifiers on structured cybersecurity datasets. Isolation Forest and Autoencoders are effective for anomaly detection, while Support Vector Machines perform well on high-dimensional data. Modern systems also integrate MITRE ATT&CK, CVE and threat intelligence feeds to enrich security events and provide context for analysts. Research in intrusion detection shows that ensemble learning algorithms such as Random Forest and XGBoost consistently outperform individual classifiers on structured cybersecurity datasets. Isolation Forest and Autoencoders are effective for anomaly detection, while Support Vector Machines perform well on high-dimensional data. Modern systems also integrate MITRE ATT&CK, CVE and threat intelligence feeds to enrich security events and provide context for analysts. Research in intrusion detection shows that ensemble learning algorithms such as Random Forest and XGBoost consistently outperform individual classifiers on structured cybersecurity datasets. Isolation Forest and Autoencoders are effective for anomaly detection, while Support Vector Machines perform well on high-dimensional data. Modern systems also integrate MITRE ATT&CK, CVE and threat intelligence feeds to enrich security events and provide context for analysts. Research in intrusion detection shows that ensemble learning algorithms such as Random Forest and XGBoost consistently outperform individual classifiers on structured cybersecurity datasets. Isolation Forest and Autoencoders are effective for anomaly detection, while Support Vector Machines perform well on high-dimensional data. Modern systems also integrate MITRE ATT&CK, CVE and threat intelligence feeds to enrich security events and provide context for analysts.

Proposed System Architecture

Security logs, vulnerability feeds and threat intelligence are collected and preprocessed. Feature engineering transforms raw data into model-ready inputs. AI models classify events, assign risk scores and send results to an interactive dashboard displaying threat trends, severity levels and security reports. Security logs, vulnerability feeds and threat intelligence are collected and preprocessed. Feature engineering transforms raw data into model-ready inputs. AI models classify events, assign risk scores and send results to an interactive dashboard displaying threat trends, severity levels and security reports. Security logs, vulnerability feeds and threat intelligence are collected and preprocessed. Feature engineering transforms raw data into model-ready inputs. AI models classify events, assign risk scores and send results to an interactive dashboard displaying threat trends, severity levels and security reports. Security logs, vulnerability feeds and threat intelligence are collected and preprocessed. Feature engineering transforms raw data into model-ready inputs. AI models classify events, assign risk scores and send results to an interactive dashboard displaying threat trends, severity levels and security reports. Security logs, vulnerability feeds and threat intelligence are collected and preprocessed. Feature engineering transforms raw data into model-ready inputs. AI models classify events, assign risk scores and send results to an interactive dashboard displaying threat trends, severity levels and security reports.

AI Models Used

Logistic Regression provides a simple baseline. Decision Tree offers explainable predictions. Random Forest improves stability through ensemble learning. XGBoost delivers excellent predictive performance for structured datasets. Support Vector Machine creates an optimal decision boundary. Isolation Forest detects unknown anomalies without labelled data. Autoencoders learn complex attack patterns through deep learning and are suitable for advanced anomaly detection. Logistic Regression provides a simple baseline. Decision Tree offers explainable predictions. Random Forest improves stability through ensemble learning. XGBoost delivers excellent predictive performance for structured datasets. Support Vector Machine creates an optimal decision boundary. Isolation Forest detects unknown anomalies without labelled data. Autoencoders learn complex attack patterns through deep learning and are suitable for advanced anomaly detection. Logistic Regression provides a simple baseline. Decision Tree offers explainable predictions. Random Forest improves stability through ensemble learning. XGBoost delivers excellent predictive performance for structured datasets. Support Vector Machine creates an optimal decision boundary. Isolation Forest detects unknown anomalies without labelled data. Autoencoders learn complex attack patterns through deep learning and are suitable for advanced anomaly detection. Logistic Regression provides a simple baseline. Decision Tree offers explainable predictions. Random Forest improves stability through ensemble learning. XGBoost delivers excellent predictive performance for structured datasets. Support Vector Machine creates an optimal decision boundary. Isolation Forest detects unknown anomalies without labelled data. Autoencoders learn complex attack patterns through deep learning and are suitable for advanced anomaly detection. Logistic Regression provides a simple baseline. Decision Tree offers explainable predictions. Random Forest improves stability through ensemble learning. XGBoost delivers excellent predictive performance for structured datasets. Support Vector Machine creates an optimal decision boundary. Isolation Forest detects unknown anomalies without labelled data. Autoencoders learn complex attack patterns through deep learning and are suitable for advanced anomaly detection. Logistic Regression provides a simple baseline. Decision Tree offers explainable predictions. Random Forest improves stability through ensemble learning. XGBoost delivers excellent predictive performance for structured datasets. Support Vector Machine creates an optimal decision boundary. Isolation Forest detects unknown anomalies without labelled data. Autoencoders learn complex attack patterns through deep learning and are suitable for advanced anomaly detection. Logistic Regression provides a simple baseline. Decision Tree offers explainable predictions. Random Forest improves stability through ensemble learning. XGBoost delivers excellent predictive performance for structured datasets. Support Vector Machine creates an optimal decision boundary. Isolation Forest detects unknown anomalies without labelled data. Autoencoders learn complex attack patterns through deep learning and are suitable for advanced anomaly detection. Logistic Regression provides a simple baseline. Decision Tree offers explainable predictions. Random Forest improves stability through ensemble learning. XGBoost delivers excellent predictive performance for structured datasets. Support Vector Machine creates an optimal decision boundary. Isolation Forest detects unknown anomalies without labelled data. Autoencoders learn complex attack patterns through deep learning and are suitable for advanced anomaly detection.

Model Performance Evaluation

Representative benchmark comparisons indicate that XGBoost achieves the highest overall accuracy followed by Random Forest. Isolation Forest performs exceptionally for anomaly detection tasks, while Logistic Regression provides fast baseline results. Evaluation metrics include Accuracy, Precision, Recall, F1-score and Training Time. Representative benchmark comparisons indicate that XGBoost achieves the highest overall accuracy followed by Random Forest. Isolation Forest performs exceptionally for anomaly detection tasks, while Logistic Regression provides fast baseline results. Evaluation metrics include Accuracy, Precision, Recall, F1-score and Training Time. Representative benchmark comparisons indicate that XGBoost achieves the highest overall accuracy followed by Random Forest. Isolation Forest performs exceptionally for anomaly detection tasks, while Logistic Regression provides fast baseline results. Evaluation metrics include Accuracy, Precision, Recall, F1-score and Training Time. Representative benchmark comparisons indicate that XGBoost achieves the highest overall accuracy followed by Random Forest. Isolation Forest performs exceptionally for anomaly detection tasks, while Logistic Regression provides fast baseline results. Evaluation metrics include Accuracy, Precision, Recall, F1-score and Training Time. Representative benchmark comparisons indicate that XGBoost achieves the highest overall accuracy followed by Random Forest. Isolation Forest performs exceptionally for anomaly detection tasks, while Logistic Regression provides fast baseline results. Evaluation metrics include Accuracy, Precision, Recall, F1-score and Training Time.

System Workflow

Collect security data, clean and normalize records, engineer features, train AI models, compare performance, predict threats, calculate risk scores and display results on an interactive dashboard with downloadable reports. Collect security data, clean and normalize records, engineer features, train AI models, compare performance, predict threats, calculate risk scores and display results on an interactive dashboard with downloadable reports. Collect security data, clean and normalize records, engineer features, train AI models, compare performance, predict threats, calculate risk scores and display results on an interactive dashboard with downloadable reports. Collect security data, clean and normalize records, engineer features, train AI models, compare performance, predict threats, calculate risk scores and display results on an interactive dashboard with downloadable reports. Collect security data, clean and normalize records, engineer features, train AI models, compare performance, predict threats, calculate risk scores and display results on an interactive dashboard with downloadable reports.

Dashboard Design

The dashboard includes KPI cards for total events, threats detected, high-risk incidents and threat percentage. Additional visualizations include confusion matrix, attack trend charts, MITRE ATT&CK mapping, severity distribution, recent alerts and filtering capabilities. The dashboard includes KPI cards for total events, threats detected, high-risk incidents and threat percentage. Additional visualizations include confusion matrix, attack trend charts, MITRE ATT&CK mapping, severity distribution, recent alerts and filtering capabilities. The dashboard includes KPI cards for total events, threats detected, high-risk incidents and threat percentage. Additional visualizations include confusion matrix, attack trend charts, MITRE ATT&CK mapping, severity distribution, recent alerts and filtering capabilities. The dashboard includes KPI cards for total events, threats detected, high-risk incidents and threat percentage. Additional visualizations include confusion matrix, attack trend charts, MITRE ATT&CK mapping, severity distribution, recent alerts and filtering capabilities. The dashboard includes KPI cards for total events, threats detected, high-risk incidents and threat percentage. Additional visualizations include confusion matrix, attack trend charts, MITRE ATT&CK mapping, severity distribution, recent alerts and filtering capabilities.

Future Scope

Future work includes SIEM integration, real-time streaming analytics, cloud deployment, generative AI assistants, automated incident response, federated learning and deep learning-based behavioural analytics. Future work includes SIEM integration, real-time streaming analytics, cloud deployment, generative AI assistants, automated incident response, federated learning and deep learning-based behavioural analytics. Future work includes SIEM integration, real-time streaming analytics, cloud deployment, generative AI assistants, automated incident response, federated learning and deep learning-based behavioural analytics. Future work includes SIEM integration, real-time streaming analytics, cloud deployment, generative AI assistants, automated incident response, federated learning and deep learning-based behavioural analytics.

Conclusion

The proposed AI-Assisted Threat Detection Dashboard demonstrates how AI can enhance cybersecurity monitoring by identifying malicious activities, reducing manual effort and improving incident response. Among the evaluated models, XGBoost and Random Forest are recommended because of their strong predictive performance and reliability. The proposed AI-Assisted Threat Detection Dashboard demonstrates how AI can enhance cybersecurity monitoring by identifying malicious activities, reducing manual effort and improving incident response. Among the evaluated models, XGBoost and Random Forest are recommended because of their strong predictive performance and reliability. The proposed AI-Assisted Threat Detection Dashboard demonstrates how AI can enhance cybersecurity monitoring by identifying malicious activities, reducing manual effort and improving incident response. Among the evaluated models, XGBoost and Random Forest are recommended because of their strong predictive performance and reliability.

Model Comparison

Model	Accuracy	Precision	Recall	F1	Remarks
XGBoost	99.1%	99.0%	99.0%	99.0%	Best overall
Random Forest	98.6%	98.5%	98.4%	98.4%	Robust

Isolation Forest	98.2%	98.0%	97.8%	97.9%	Best anomaly
SVM	97.3%	97.2%	97.1%	97.1%	High dimensional
Decision Tree	96.8%	96.5%	96.4%	96.4%	Explainable
Logistic Regression	93.8%	93.5%	93.4%	93.4%	Baseline

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